Combating Popularity Bias in Graph-Based Collaborative Filtering: Interventions for LightGCN

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1 Motivation and Background

1.1 The Problem

Personalized recommendation systems are some of the most widely used applications of graph machine learning. These applications represent user-item interactions as a bipartite graph and use GNNs to predict which unseen items a user is likely to prefer. LightGCN [He et al., 2020] is one particular lightweight skeleton for this type of problem.

Neural Graph Collaborative Filtering (NGCF) [Wang et al., 2019] was among the first to bring GCNs to the field of recommendation systems. They used the standard recipe for GCN propagation, which involves neighborhood aggregation and a feature-transformation matrix + nonlinear activation at each of its layers. This follows the design of GCNs that were originally built for node classification on graphs with attributes.

[He et al., 2020] asked the question: is this recipe actually appropriate for collaborative filtering (predicting a user's interests by collecting/analyzing the preferences from other users)? It's important to understand that in this case, the graph is not feature-rich from content and is instead *only* the edges between user IDs and item IDs. This means that vector embeddings must be learned solely from the interaction graph. By doing targeted ablation studies on the original NGCF, they found that the feature-transformation matrices and nonlinear activation were not pulling their weight in improving the performance of the output. By removing these from the architecture, they actually saw a large improvement over NGCF.

Furthermore, they showed that this was not just a case of overfitting, but actually a training-difficulty issue: the extra operations that the model was making made it genuinely harder for it to optimize well. This was the motivation behind LightGCN, which keeps just the normalized neighborhood aggregation and a weighted combination of embeddings across its layers.

1.2 Popularity Bias Persists Despite Simplification

The simplification from NGCF to LightGCN properly addresses this training-difficulty problem, but it does not address a separate issue: *popularity bias*. During propagation, an item's embedding is updated as a function of its neighboring users. As a result of this, the items with many interactions end up in far more aggregation steps than other items, so their embeddings get reinforced every layer regardless of whether a given user would genuinely prefer them [Klimashevskaia et al., 2024].

Since this bias is part of the graph convolution itself rather than by any of the components that were removed to create LightGCN, the simplification of the architecture doesn't do anything to correct it. In fact, a more recent paper has shown that LightGCN behaves inconsistently across the graph in ways that can compound this bias even further [Lee et al., 2024].

Preprint.

2 Proposed Approach

My contribution will be not one new architecture but a mix-and-match of three intentional tweaks to LightGCN, each combating the popularity bias issue at a different stage along the LightGCN pipeline. I will ablate each to determine their sole effect on the model's behavior. I will use MovieLens-1M [Harper & Konstan, 2015], a new benchmark that was not used for LightGCN. I will use this benchmark because it contains not only the interaction graph information, but also item content metadata such as genre and release year, which is required for the content-based cold-item initialization I plan to add as an intervention.

2.1 Baseline: LightGCN propagation

My baseline adopts LightGCN's propagation and layer-combination scheme [He et al., 2020], unmodified and trained with Bayesian Personalized Ranking (BPR) loss [Rendle et al., 2012]. I plan to leave the propagation step itself unchanged.

2.2 Intervention A: Degree-Discounted Aggregation

For this tweak, I will introduce an additional, tunable discount on high-degree (popular) items during aggregation:

$$e_i^{(k+1)} = \sum_{u \in \mathcal{N}(i)} \frac{1}{\sqrt{|\mathcal{N}(u)||\mathcal{N}(i)|}} \cdot |\mathcal{N}(i)|^{-\gamma} \, e_u^{(k)}, \qquad \gamma \in [0,1], \tag{1}$$

which reduces the rate at which popular items' embeddings are reinforced across successive propagation layers via the popularity as a negative exponent. This means that the more popular an item is, the less the embedding gets updated during this step. If we set $\gamma = 0$, we turn off this intervention entirely (reverting to base LightGCN). If we set it to $\gamma = 1$, we are using the maximum popularity discount.

2.3 Intervention B: Popularity-Aware Loss Re-weighting

This interventions will re-weight the BPR loss in a way that decreases the weight of popular positive items towards loss during training:

$$\mathcal{L} = -\sum_{(u,i,j)} w(i) \cdot \ln \sigma(y_{ui} - y_{uj}), \qquad w(i) = \mathrm{pop}(i)^{-\beta}, \quad \beta \in [0,1] \tag{2}$$

This ensures that the gradient signal is not dominated by a small set of frequently-interacted items, and the signal of rare items can still have an effect on training.

2.4 Intervention C: Content-Based Cold-Item Initialization

Instead of purely relying on the graph interactions to help accumulate meaning for an item's embedding, we create a small function that initializes items based on their metadata to give them a starting position in the vector space that "makes sense":

$$e_i^{(0)} = e_i^{\mathrm{ID}} + f_\theta(x_i) \tag{3}$$

where x_i is a feature vector built from genre one-hot encoding and normalized release year, and f_θ is a lightweight MLP. This way, even an item that nobody has interacted with doesn't just have a completely random or meaningless embedding.

3 Experimental Plan

I will implement LightGCN and all three interventions on top of it myself (no pre-existing LightGCN black box will be used). I will compare base LightGCN as a control against all three interventions both individually and in all combinations on various splits of MovieLens-1M, which contains roughly 1 million ratings from 6,040 users on 3,900 movies [Harper & Konstan, 2015], which I will binarize into implicit feedback.

For each configuration, I will report Recall and NDCG (ranking accuracy), Coverage (share of catalog recommended), Average Recommendation Popularity, and Gini coefficient (bias/exposure inequality). My central analysis is the accuracy-bias trade off (Recall vs. Gini) across all configurations, testing whether the combined method, any single intervention, or the base LightGCN is best at reducing bias while maintaining accuracy.

Computational resources. LightGCN is deliberately lightweight, so I expect the full ablation grid to run comfortably on either a Google Colab T4 instance or on Kaggle's notebook execution environment. I expect experiments to run for a few hours of GPU time.

Key deliverable. As mentioned above, the primary goal of this project is to analyze the Recall vs. Gini tradeoff for the control (base LightGCN) and ablations/combinations of the three proposed interventions. My hypothesis is that the combined intervention will achieve a better tradeoff, which would signal that it reduces bias in comparison to base LightGCN.

References

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